

Project: Task 7 - Logistic Regression on Titanic Dataset Model: Logistic Regression

1. Data Preprocessing

Missing Values: Imputed Age with the median and Embarked with the mode.

Encoding: Applied One-Hot Encoding to Sex and Embarked.

Scaling: Applied Standard Scaler to Age and Fare to ensure the model converges efficiently.

2. Model Performance

Accuracy: ~81% (The model correctly predicted survival for 81% of passengers).

Recall: ~73% (The model correctly identified 73% of the people who actually survived).

F1-Score: ~0.76 (A balanced score between Precision and Recall).

3. ROC & AUC Analysis

AUC Score: ~0.88

Interpretation: An AUC score of 0.88 indicates the model has a strong ability to distinguish between survivors and non-survivors. The ROC curve stays well above the diagonal line, confirming good performance.